

Section 3.1

I. Secant Line through $(x_0, f(x_0))$ to $(x_0 + \Delta x, f(x_0 + \Delta x))$

Secant Line through $(x_0, f(x_0))$ to $(x_1, f(x_1))$

- a. $\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$
- b. Difference Quotient
- c. Slope of Secant line
- d. Average Rate of Change of f over the interval $[x_0, x_0 + \Delta x]$

II. Tangent Line = Limiting Case of Secant Line through $(x_0, f(x_0))$ to $(x_0 + \Delta x, f(x_0 + \Delta x))$

Tangent Line = Limiting Case of Secant Line through $(x_0, f(x_0))$ to $(x_1, f(x_1))$

- a. $\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(x_0)$
- b. Derivative = $\lim_{\Delta x \rightarrow 0}$ Difference Quotient
- c. Slope of Tangent Line = $\lim_{\Delta x \rightarrow 0}$ Slope of Secant Line
- d. Instantaneous Rate of Change of f at $x_0 = \lim_{\Delta x \rightarrow 0}$ Average Rate of Change of f over the interval $[x_0, x_0 + \Delta x]$

Examples

III. Definition of Derivative of f at x

Examples

- IV. Equation of Tangent Line to a curve $y = f(x)$ at $x = x_0$
Equation of Normal Line to a curve $y = f(x)$ at $x = x_0$

Examples

- V. Relationship between f and f'

- VI. Existence of Derivative

- VII. Differentiability and Continuity